

How We Tested

The Test Rig

Our test machine is fairly entry-level—an Intel Pentium 4 560 3.2 GHz processor on a Gigabyte GA-965P-DQ6 motherboard with 1 GB of 533 MHz DDR2 RAM, an NVIDIA GeForce 7600GT graphics card, and a 120 GB 7200 rpm ATA IV Seagate Barracuda hard disk. We used Windows XP with SP2 as the OS.

Media used

Text/combi-document printouts: 100 gsm (grams per square metre) paper from Berga (A4 size)

Photo printouts: 160 gsm glossy photo paper from Novajet (A4 size)

Features

We look for features such as PictBridge, memory card readers, fax, input tray capacity, number of cartridges, and so on for the device in general. For the scanner, we consider the maximum scan resolution, bit depth, scanner type, and the presence of automatic document feeders (ADF). For the copier aspect, we look for the ability to work as a standalone device, the maximum number of pages that can be printed in multi-copy mode, reducing or enlarging a copy, and more.

The Printer Tests

We use a text document to test the raw print speeds of the printers of the MFDs. To test the ability of the MFDs to handle the various aspects of a regular document, we create a combi-document with a mixture of black text, graphs, and images. We

test print quality at Normal and Best quality for the combi-document, and only Best for photo prints. We also clock the time taken in each case.

The Scanner Tests

To test the scanners for speed, we scanned an A4-size image five times and average the time taken for each scan—the scanners start out cold, so we need to compensate for the warm-up time.

We imported the same image at 150 dpi and 600 dpi in Adobe Photoshop CS2, and scanned a full text document in black-and-white at 200 dpi. The scanners then underwent a test to determine colour differentiating abilities using the Kodak IT8 card with different shades of colour—each a little different from the preceding one. Next, we used the resolution chart—a chart with five boxes with parallel lines packed very close, and each subsequent box with lines packed more densely. We scanned the resolution chart at 300 dpi. This test gives us a measure of the scanner's capability to capture minute details—whether it's able to discern between the lines or not. Last was an OCR test to check the scanner's ability to differentiate light and dark areas: a print article was scanned into ABBYY Fine Reader Pro 8.0 PE, and checked for wrongly - interpreted letters. We counted the number of mis - spelled letters in each case.

The Copier Tests

Here we copied the text document in black-and-white, and the combi-document in colour, and rated the MFDs on their speed and print quality.

the HP MFDs seemed to have problems distinguishing between adjacent shades in the light and the dark colours. The rest produces scans of acceptable quality.

The copy quality of the Canon is the best, followed by the HPs. All in all, none of the MFDs are particularly disappointing, and copies are good enough to save you a trip to the photocopy centre in a crunch situation.

OFFICE MFDs

MFDs have always been used more in offices, where space-saving is important. It is therefore no wonder that half the number of MFDs we've tested fall under the Office MFD category.

All falling in the price range of Rs 6,000 to Rs 10,000, nine MFDs feature in this segment: two from Brother (the DCP-330C and MFC-240C), the Canon PIXMA MP460, four from HP (the Officejet 4355 All-in-One, Officejet 5610 All-in-One, Officejet 5680 All-in-One, and Photosmart C4188 All-in-One), and the Lexmark X4270.

Features Printing

The HP and Lexmark MFDs have two ink cartridges, while the

Brother, Canon, and Epson have four each. The input paper tray capacity for all of them is 100, and this is more than enough for a moderate-size office with moderate workloads.

Scan, Copy, And Fax

For the price, the HP Officejet 4355 and Lexmark X4270 pack in a lot of features. The 4355 boasts of a good scan resolution of 1200 x 1200 dpi. The Lexmark X4270 is a direct competitor, but has a lower scan resolution of 600 x 1200 dpi. The gripe in both cases is that these feature a sheet-fed scanning element and are therefore slower than their flatbed counterparts. There is



HP Photosmart C4188 All-in-One
Performance matches features

also the chance of documents getting crumpled inside the scanner. Fax, complete with handset, is also part of the parcel in these two MFDs. The HP 4355 can create up to 200 copies, the largest number in this entire test.

The HP C4188 and 5610 feature high-resolution scanning at 1200 x 2400 dpi, enabling them to pick up even fine print from documents. The Brother DCP-330C and HP C4188 have flatbed scanners, while the HP 5610 has the option of a sheet-fed scanner too. A sheet-fed scanner, in conjunction with an automatic document feeder (ADF), relieves you from the task of feeding a number of pages to be scanned—simply stash the paper in the ADF and the device will automatically pick the papers one by one, and scan them in proper order.

The Epson MFD doesn't have fax, and the Canon PIXMA MP460 has neither fax nor ADF.

Build Quality And Other Features

The HP Officejet 4355 and Lexmark X4270 are extremely compact. They are built ruggedly overall, but the trays that protrude out of them are a little